Due Thursday November 6, 11:59pm PHYS 2210 Fall 2025- Dr. De Grandi

Homework 9 - Part 2

Instructions: To guarantee full credit **show all your steps**, if you are using some additional equations that are not the ones we discussed in class (not on the formula sheet) you need to be able to derive them. Follow the Tips for Homework Problem Solving. To submit your homework correctly on Gradescope, follow the instructions on the Homework Format And Submission page.
If you have any questions post them on the Homework 9- Questions discussion board or come to Study-hall.

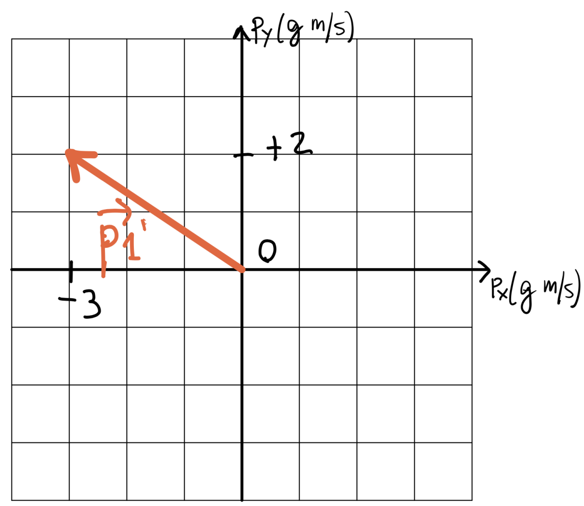
Problem 1: Bumblebees collision

Hints! Review the 2D collision example discussed at the end of Lecture 17 (find Notes and Slides under Modules: Week 10). Practice first with Questions 1 and 2 of Part 1 of the homework.

Bumblebees are known to collide with objects regularly. Their body is such that they experience no injuries and can quickly readjust their flight and continue their way.
Two bumblebees are flying towards each other: bumblebee 1 is moving horizontally to the *right* (positive x-direction in the figure below) with a speed of 10 m/s, bumblebee 2 is moving to the *left* (negative x-direction in the figure below) with a speed of 20 m/s. The two bumblebees collide and bounce off each other. Each bumblebee has a mass $m = 0.1$g (where g=grams=10^{-3} kg).

The figure below shows the momentum vector $\mathbf{p}_1'$ of bumblebee 1 *after* the collision.

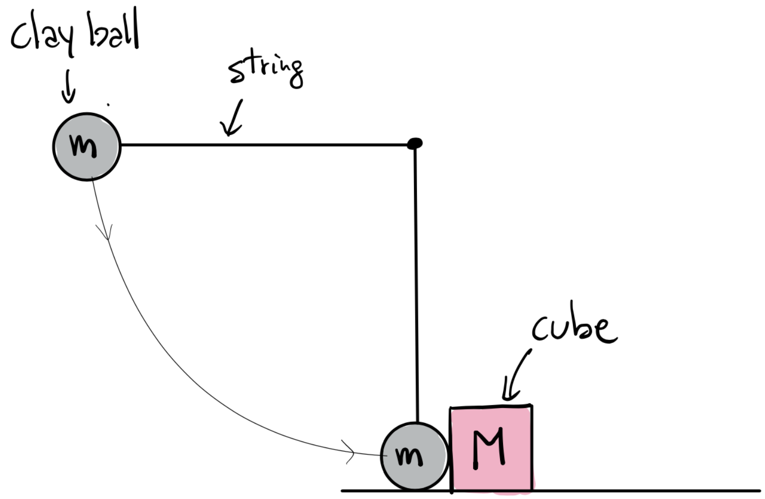
1. Calculate the momentum vector $\mathbf{p}_2'$ of bumblebee 2 *after* the collision.
2. **Show clearly ALL the steps of your calculation and reasoning.**
3. Write out the components of the vector.
4. Draw the vector on the figure below.



Problem 2: Sticky collision

Hints! Review the solutions to Disc. Section 15 *Inelastic Collision* (find it under Modules: Week 10).

A clay ball of mass m is attached to a string of length L and fixed at the other end. The ball is released from rest when the string is horizontal (as shown in the figure below). At the bottom of its swing the ball collides with a plastic cube of mass M initially at rest on a frictionless surface.
Assume the clay ball sticks to the cube after the collision (hence it's a perfectly inelastic collision) and that the system of ball+cube swings up to the right.



a) What is the maximum height (above the surface) that the system of ball+cube will reach after the collision? Show the steps of your calculation and reasoning and make sure your answer is in terms of the given quantities only (*Tip*: Use dimensional analysis to check your answer has the correct dimension for a height).

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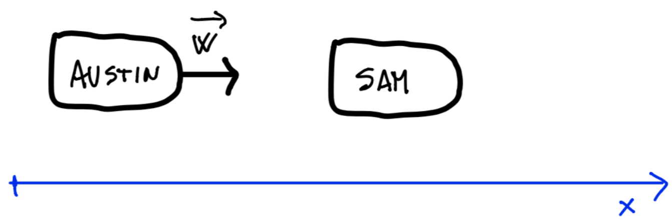
b) What is the angle that the string will make with the vertical when the system of ball+cube reaches its maximum height? Find an expression in terms of the given quantities.

Problem 3: Austin and Sam on bumping cars

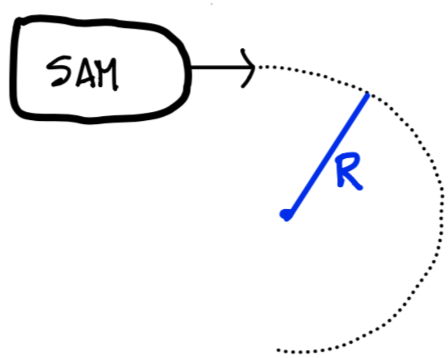
Austin and Sam are driving bumping cars at the amusement park and observing the effects of elastic collisions. Sam is temporarily stopped, when Austin comes straight behind him with a speed w and bumps his car from the back. The collision can be considered completely elastic and the motion occurs along one direction, along an x-axis as shown in the figure below. Austin's car overall mass is m_A and Sam's car overall mass is m_S.

Hints! Revise the notes of Lecture 17 (and the formula sheet) for the equations applicable to elastic collisions in 1 dimension. **You do not have to derive those equations, you can just use them.**



a) Find an expression for Sam's car velocity after the collision in terms of the given quantities. Make sure to specify both the magnitude and direction. Please show all the steps of your reasoning (if you use any formula for collisions from our formula sheet make sure to explain how you applied it to this situation).

b) Right after the collision Sam wants to take a sharp turn (i.e. perform half a circle), he knows that friction will be the only force keeping him from sliding off the curve, so he needs to be careful in choosing the radius of the curve not to be too tight. Assume the coefficient of static and kinetic friction with the floor are μ_s and μ_k, respectively. What is the minimum curve radius that Sam can take without sliding off the curve? Show the steps of your reasoning and calculation and find an expression for the minimum safe radius in terms of the given quantities only.



c) How does the minimum radius you found in the previous part depend on Austin's initial speed w? How would this minimum radius change if Austin had bumped Sam at twice the speed?